

Koopman Autoencoders Uncover Interpretable Eigenmodes in LLM Latent Dynamics

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Abstract

Autoregressive large language models (LLMs) have been extensively studied through static representational analyses, yet the dynamics of their token-wise activation trajectories remain insufficiently characterized. Extending from Latent Spectral Dynamics, which establishes a stochastic autonomous dynamical systems framework on windowed residual activations, we refine the data-constrained and less-interpretable Hankel-EDMD method with Koopman Autoencoders (KAEs) – deep encoder-decoder networks that learn nonlinear observable bases and a finite-dimensional linear Koopman operator simultaneously. Applied to residual stream trajectories from multiple language models across mathematical reasoning and narrative generation tasks, the KAEs successfully fit the generation dynamics with low reconstruction errors, yielding three principal findings: (1) an *invariant eigenspectrum* at fixed latent dimension that persists across datasets and model families; (2) a *resolution scaling* phenomenon when increasing the encoder latent dimension progressively reveals finer transient structure while preserving previously recovered modes; and (3) an *interpretability pipeline* – inspired by sparse autoencoder feature analysis – that taxonomizes learned Koopman eigenmodes into a structural–transient hierarchy. Structural components (high-energy backbone and oscillatory modes) provide the document’s scaffolding, tracking broad formatting boundaries and problem-to-answer transitions. Conversely, the spectrum is dominated by low-energy transient modes that capture localized events. Such modes mostly encode turn delimiters, while a smaller subset fires during discrete computational steps like equation-to-result pivots. Linear probing confirmed these labels’ internal consistency, while causal ablation showed that perturbing specific modes shifted outputs significantly more than random baselines. These results suggest that autoregressive LLM generation admits a spectral decomposition into a dominant structural backbone and sparse reasoning-specific transients, strengthening a dynamical-systems foundation for interpretability.

1 Introduction

Autoregressive large language models (LLMs) are increasingly deployed as autonomous agents capable of advanced reasoning, yet their internal computational processes remain opaque. Mechanistic interpretability has produced powerful tools for dissecting Transformer internals—circuit analysis (Elhage et al., 2021), probing methods (Belrose et al., 2023), representation engineering (Zou et al., 2023), and sparse autoencoders (Wang et al., 2024)—but these approaches operate on *static* snapshots: fixed layers, token positions, or weight states. Characteristics of the *temporal evolution* of activations across the token-by-token generation process, such as how the residual stream transitions between computational phases, oscillates during uncertainty, or converges toward an answer, are invisible to methods that analyze individual snapshots in isolation.

A recent line of work has begun to address this gap by framing neural computation through a dynamical-systems lens. Deep Equilibrium Models (Bai et al., 2019) and Neural ODEs

(Chen et al., 2018) established that neural computation can be analyzed as a dynamical system.

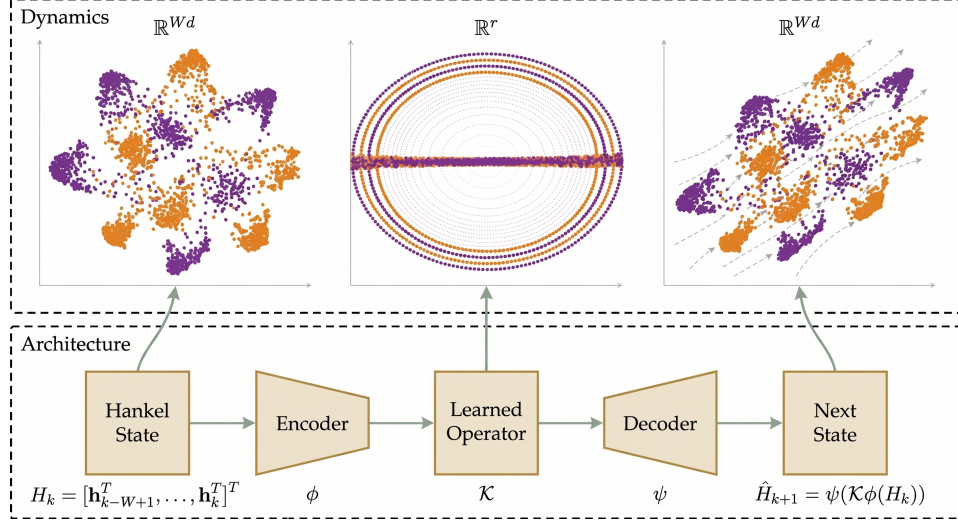


Figure 1: Koopman Autoencoder overview. **Bottom:** the encoder ϕ maps Hankel states $H_k \in \mathbb{R}^{Wd}$ to a latent space $\mathbb{R}^r$, where the learned operator $\mathcal{K}$ advances states linearly, and the decoder ψ reconstructs the next Hankel state. **Top:** the corresponding dynamics—nonlinear trajectories in ambient space (left, right) become near-circular orbits under the linear operator in latent space (center).

The Latent Spectral Dynamics (LSD) framework (Anonymous, 2026) formalized autoregressive LLM generation as a stochastic, autonomous, discrete-time dynamical system on time-delay-embedded residual stream states, and applied Koopman operator theory via Hankel-Extended Dynamic Mode Decomposition (Hankel-EDMD) to extract spectral signatures of truthful versus untruthful generation. That work demonstrated that truthful trajectories converge toward a stable invariant measure with near-unity real eigenvalues, while untruthful trajectories exhibit oscillatory dynamics driven by complex-conjugate eigenvalue pairs—a distinction invisible to static representational analyses.

However, the Hankel-EDMD approach faces critical scaling limitations. Its reliance on exact matrix factorizations over massive, uncompressed coordinate bases—with Gram matrices of dimension $N \times N$ and ambient observable dimensions $D = Wd$ exceeding 25,000—makes it prohibitively expensive when scaling to larger models or longer generation contexts. Furthermore, the coordinate basis is restricted to linear observables, potentially missing nonlinear dynamical structure in the residual stream.

In this work, we address both limitations by replacing Hankel-EDMD with Koopman Autoencoders (KAEs): deep encoder-decoder networks that jointly learn a nonlinear observable basis and a finite-dimensional linear Koopman operator. This substitution achieves *computational scalability* by bypassing cubic-cost matrix factorization in favor of stochastic gradient descent, and *nonlinear observable learning* that captures dynamics invisible to linear bases. We apply this framework to residual stream trajectories from three 14B-parameter LLMs across mathematical reasoning (GSM8K, MATH500) and open-text generation (WikiText-103), and develop a systematic *interpretability pipeline* for the learned Koopman eigenmodes.

Our main contributions are:

1. **Scalable Operator Learning.** We introduce KAEs as a deep learning extension to Hankel-EDMD for approximating the Koopman operator on LLM residual stream trajectories, bypassing the cubic-cost matrix factorization that limited the prior framework (Section 3).

2. **End-to-end Interpretability Pipeline.** Adapting sparse autoencoder (SAE) feature analysis to the dynamical setting, we design a five-step pipeline: eigenmode taxonomy, max-activating segment extraction, LLM-assisted labeling, linear probing, and causal ablation (Section 5.4).
3. **Spectral Analysis Discovery: Invariant Spectra and Resolution Scaling.** KAEs trained independently across datasets and model families recover similar spectral geometries at fixed r , suggesting a universal autoregressive backbone; increasing r from 64 to 512 progressively reveals finer transient structure while preserving recovered modes (Sections 5.2–5.3).
4. **Interpretability Discovery: Structural–Transient Hierarchy.** We discover that high-energy backbone and oscillatory eigenmodes track the structural skeleton of generation—problem boundaries, formatting templates, turn transitions—while low-energy transient eigenmodes capture brief, reasoning-specific events such as equation-to-result pivots (Section 5.4).

2 Dynamical Systems Formulation

We briefly review the dynamical formulation of [Anonymous \(2026\)](#), which provides the necessary theoretical foundation for our KAE extension.

2.1 Defining Key Notations

A discrete-time deterministic dynamical system $(\mathcal{X}, F)$ evolves states $x_{k+1} = F(x_k)$ on a state space $\mathcal{X}$ via a deterministic transition map $F : \mathcal{X} \rightarrow \mathcal{X}$, generating a trajectory $(x_0, x_1, x_2, \dots)$ from initial condition x_0 . To analyze nonlinear trajectories such as LLM token-wise activations, we rely on Koopman operator theory: an *observable* is any scalar measurement $g : \mathcal{X} \rightarrow \mathbb{C}$ of the state, and the *Koopman operator* $\mathcal{K}$ advances observables linearly forward in time via $(\mathcal{K}g)(x_k) = g(F(x_k))$. This lifts nonlinear state dynamics into a linear evolution on functions, enabling spectral decomposition of opaque generation behavior.

2.2 Constructing Dynamical States

At generation step k , an autoregressive Transformer maintains a KV cache S_k that acts as an exact Markov state: $P_\phi(x_{k+1}|x_{1:k}) = P_\phi(x_{k+1}|S_k)$, where ϕ denotes the model parameters. Because S_k is prohibitively high-dimensional for operator estimation, we construct a fixed-dimensional surrogate via Hankel time-delay (windowed) embedding of the *last-layer residual stream* $h_k \in \mathbb{R}^d$:

$$H_k = [h_{k-W+1}^\top, \dots, h_k^\top]^\top \in \mathbb{R}^{Wd} \quad (1)$$

By the measure-theoretic phase-space reconstruction theorem of [Botvinick-Greenhouse et al. \(2025\)](#), a sufficient window size W ensures that the delay-embedded state H_k preserves the statistical structure of the true state, meaning $P_\phi(x_{k+1}|H_k) \approx P_\phi(x_{k+1}|S_k)$. Throughout this paper, the surrogate H_k serves as the core dynamical state modeled by KAEs.

2.3 Autonomous Dynamics and the Koopman Operator

From the original LSD framework, under greedy decoding ($\tau = 0$), which all experiments in this paper adopt, one obtains a discrete-time deterministic dynamical system

$$H_{k+1} = F_{\phi, \theta^*}(H_k) \quad (2)$$

where F is the transformer forward pass, ϕ denotes the fixed model parameters, and θ^* is the prompting regime (e.g. truthful). An *observable* is any scalar measurement of the state, $g : \mathbb{R}^{Wd} \rightarrow \mathbb{C}$. The *Koopman operator* $\mathcal{K}_{\theta^*}$ advances observables forward by one timestep, lifting the nonlinear state dynamics into a linear evolution on functions:

$$(\mathcal{K}_{\theta^*}g)(H_k) = g(F_{\phi, \theta^*}(H_k)) \quad (3)$$

Notably, LSD approximated $\mathcal{K}_{\theta^*}$ via Hankel-EDMD with Tikhonov regularization, which faces computational constraints.

3 Koopman Autoencoders for LLM Latent Dynamics

Seeking a computationally scalable and interpretable alternative for approximating $\mathcal{K}_{\theta^*}$, we replace the traditional Hankel-EDMD framework with a Koopman Autoencoder.

3.1 Architecture

A Koopman Autoencoder consists of three components:

- An **encoder** $\phi : \mathbb{R}^{Wd} \rightarrow \mathbb{R}^r$ serving as the learned observable dictionary,
- A **learnable matrix** $K \in \mathbb{R}^{r \times r}$ approximating the Koopman operator in the latent space,
- A **decoder** $\psi : \mathbb{R}^r \rightarrow \mathbb{R}^{Wd}$ reconstructing Hankel states from latent representations.

The system is trained by minimizing a combined objective:

$$\mathcal{L} = \underbrace{\|H_k - \psi(\phi(H_k))\|^2}_{\text{Reconstruction}} + \alpha \underbrace{\|\phi(H_{k+1}) - K\phi(H_k)\|^2}_{\text{Linear dynamics}} + \beta \underbrace{\sum_{n=1}^N \|\phi(H_{k+n}) - K^n \phi(H_k)\|^2}_{\text{Multi-step rollout}} \quad (4)$$

where α and β are tunable parameters. Specifically, the reconstruction term prevents degenerate encodings, the linear dynamics term enforces that transitions are linear in the learned latent space, and the rollout term stabilizes eigenvalues over longer horizons.

We design our KAE to operate directly on the full Hankel state $H_k \in \mathbb{R}^{Wd}$ without prior dimensionality reduction. The architecture utilizes a two-layer MLP encoder ($\mathbb{R}^{Wd} \rightarrow \mathbb{R}^{128}$ with $\text{ReLU} \rightarrow \mathbb{R}^r$) and a symmetric decoder. For $W = 8$ and $d = 5120$, the input dimension is $Wd = 40,960$.

3.2 Spectral Analysis Setup

The learned matrix K admits eigendecomposition $K = V\Lambda V^{-1}$, verified empirically via the condition number $\kappa(V) \in [194, 328]$ across all experiments (Appendix A). The decomposition yields three objects:

Eigenvalues. Each $\lambda_j \in \mathbb{C}$ encodes the temporal signature of mode j : the magnitude $|\lambda_j|$ governs decay rate (e.g. $|\lambda_j| < 0.5$: fast-decaying *transient*; $|\lambda_j| \geq 0.9$: persistent), and the phase $\arg(\lambda_j)$ governs oscillation frequency (near-zero phase: monotonic *backbone modes*; nonzero phase, typically in conjugate pairs: *oscillatory modes*).

Modal coefficients. $c_j(k) = (V^{-1}\phi(H_k))_j \in \mathbb{C}$ measures the activation of mode j at token k , evolving as $c_j(k+n) \approx \lambda_j^n c_j(k)$.

Koopman modes. The operator $K = V\Lambda V^{-1}$ defines Koopman modes as the latent eigenvectors $\{v_j\}_{j=1}^r$. The encoder output $z_k = \phi(H_k)$ decomposes as

$$c(k) = V^{-1}z_k \in \mathbb{C}^r, \quad z_k = \sum_{j=1}^r c_j(k) v_j. \quad (5)$$

The modal amplitudes $|c_j(k)|$ identify when each spectral mode is active along a trajectory, and the decoder ψ maps latent states back to Hankel space.

Together, the eigenvalues carry the temporal structure (frequency, decay rate) while the Koopman modes carry the spatial structure (which latent dimensions participate), grounding the interpretability analysis in Section 5.4.

4 Experimental Setup

Models. We use three 14B-parameter instruction-tuned LLMs: Qwen2.5-14B-Instruct, DeepSeek-R1-Distill-Qwen-14B, and Mistral-3-14B.

Datasets. For each model, we extract final-layer residual stream trajectories ($d = 5120$) from 500 sequences in each of three domains: (i) **GSM8K**, grade-school math word problems requiring multi-step arithmetic; (ii) **MATH500**, competition-level mathematics with heavier LaTeX formatting; and (iii) **WikiText-103**, open-text continuation serving as a non-reasoning control. All sequences are capped at 512 tokens under greedy decoding ($\tau = 0$).

KAE Configuration. We train a nonlinear KAE with Hankel window $W = 8$ and sweep the latent dimension across $r \in \{64, 128, 256, 512\}$, using a multi-step rollout horizon of $N = 8$. Training uses AdamW with learning rate 10^{-4} and batch size 256, with an 80/10/10 train/validation/test split stratified by trajectory. We evaluate fit quality via the multi-step rollout MSE:

$$\text{MSE}(n) = \frac{1}{N_n} \sum_{i=1}^{N_n} \|H_{k_i+n} - \psi(K^n \phi(H_{k_i}))\|_2^2, \quad (6)$$

which measures the reconstruction error of the learned operator at horizon n , computed on held-out test trajectories. Full hyperparameters are listed in Appendix A.

5 Results

5.1 Dynamical Fit Quality

Table 1 reports rollout MSE across horizons and datasets for Qwen2.5-14B. The stable degradation from 0.478 (1-step) to 0.727 (8-step) on GSM8K confirms that the learned operator captures the dominant Markovian transition structure. All three datasets converge with validation losses below 1.35 and spectral radii $\rho(K) \approx 1.00$ –1.01 (Appendix A, Table 8).

Table 1: KAE unnormalized rollout MSE across datasets (Qwen2.5-14B-Instruct, $r = 256$, $W = 8$). MSE is measured in standardized Hankel space on train/val/test splits (400/50/50).

Horizon	GSM8K			MATH-500			WikiText-103		
	Train	Val	Test	Train	Val	Test	Train	Val	Test
1-step	0.406	0.448	0.478	0.488	0.595	0.538	0.499	0.534	0.522
4-step	0.418	0.512	0.546	0.502	0.673	0.617	0.512	0.581	0.572
8-step	0.476	0.686	0.727	0.597	0.870	0.821	0.573	0.679	0.676

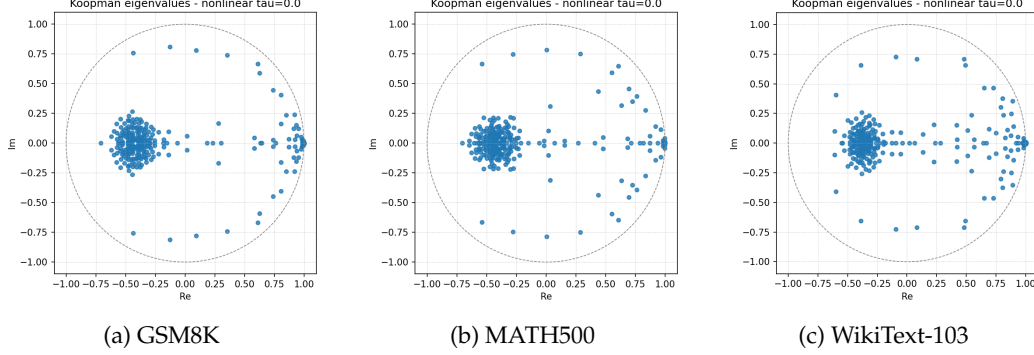
Comparison to EDMD baseline. On the truthful/untruthful regime of [Anonymous \(2026\)](#), the nonlinear KAE ($r = 64$, $W = 5$) achieves 1-step test MSE of 0.525 (untruthful) and 0.636 (truthful), degrading to 0.871 and 0.896 at $n = 8$. The coordinate EDMD baseline yields relative MSE of 0.22→0.75 (untruthful) and 0.24→0.79 (truthful) over the same horizons—a much steeper degradation. Because the two methods use different coordinate systems (variance-normalized vs. raw Hankel space), absolute MSE values are not directly comparable, but the multi-step stability gap consistently favors the nonlinear KAE.

5.2 Invariant Eigenspectra Across Datasets

At fixed $r = 256$, KAEs trained separately on GSM8K, MATH500, and WikiText-103 yield strikingly similar spectral geometries (Figure 2). All three spectra share: (i) a dense near-unity real cluster (backbone modes); (ii) a negative-real cluster near $\text{Re} \approx -0.4$ (period-2 oscillations); and (iii) scattered complex-conjugate pairs throughout the unit disk.

Because each KAE learns its own nonlinear observable basis, these operators inhabit different latent spaces, making direct numerical comparison ill-posed; quantitative confirmation would require coordinate-free invariants or shared-encoder experiments. Nevertheless, the persistence across domains with very different token-level statistics is suggestive of a universal autoregressive dynamical backbone.

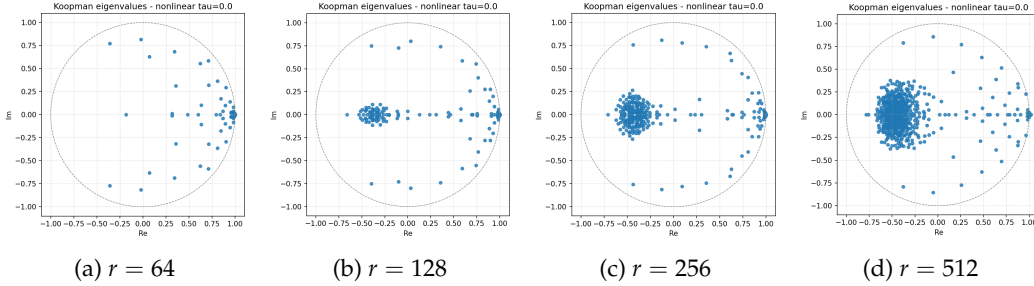
Figure 2: Eigenvalue spectra of the learned Koopman operator at $r = 256$, $W = 8$ (Qwen2.5-14B-Instruct). All three share the same structure: a cluster near $+1$ on the real axis (right), a cluster near -0.4 (left) corresponding to period-2 oscillations, and scattered complex-conjugate pairs. Dashed circle: $|\lambda| = 1$.



5.3 Resolution Scaling of Eigenspectra

Figure 3 shows eigenspectra at $r \in \{64, 128, 256, 512\}$ on GSM8K. At $r = 64$, the near-unity backbone is already present but complex modes are sparse. As r increases, a dense cluster progressively forms around $\text{Re} \approx -0.4$, while the backbone persists stably. This resolution scaling replicates across all three models (Appendix C).

Figure 3: Resolution scaling of the Koopman eigenspectrum (Qwen2.5-14B-Instruct, GSM8K, $W = 8$). The near-unity backbone persists across all r , while a dense period-2 cluster emerges progressively. Dashed circle: $|\lambda| = 1$.



5.4 Interpreting the Structural-Transient Mode Hierarchy

Inspired by the Sparse Autoencoder (SAE) analysis framework, we develop a five-step interpretability pipeline applied to all 256 modes across all three datasets.

5.4.1 Step 1: Eigenmode Taxonomy

We begin by sorting the 256 learned modes into coarse functional classes using their spectral properties. For each mode j , we record eigenvalue magnitude $|\lambda_j|$, phase $\theta_j = \arg(\lambda_j)$, and empirical energy fraction E_j (the normalized cross-trajectory variance of $|c_j(k)|^2$; see Appendix B for the full definition). Using quantile-based thresholds on $\{E_j\}$, modes are classified as: *transient* ($|\lambda_j| < 0.5$ or very low energy), *backbone* (high $|\lambda_j|$, near-real phase, high energy), or *oscillatory* (all remaining). Note that the cutoff is heuristic and only serves as an initial classification, with finer execution left for future work.

Key result. The three-tier hierarchy is consistent across all datasets (Table 2). Transient modes comprise $\sim 63\%$ of all modes but carry less than 20% of the energy. In MATH500, 16 backbone modes absorb 99.0% of the total energy, while WikiText-103 has the most backbone modes (35), reflecting greater document-level diversity. Eigenvalue magnitude ranges per class are in Appendix B.

Table 2: Eigenmode taxonomy at $r = 256$, $W = 8$ (Qwen2.5-14B-Instruct). n : mode count; $\sum E_j$: cumulative energy fraction (share of total activation variance explained by that class); $|\lambda|$: mean eigenvalue magnitude.

Class	GSM8K			MATH500			WikiText-103		
	n	$\sum E_j$	$ \lambda $	n	$\sum E_j$	$ \lambda $	n	$\sum E_j$	$ \lambda $
Backbone	23	0.567	0.988	16	0.990	0.991	35	0.486	0.990
Oscillatory	73	0.244	0.717	72	0.008	0.687	58	0.311	0.748
Transient	160	0.189	0.397	168	0.001	0.391	163	0.204	0.354

5.4.2 Step 2: Max-Activating Segments

With modes classified, we next ask *where* each mode fires in the generated text. For each mode j , we identify high-activation positions by thresholding $|c_j(k)|$ at $\mu_j + 1.5\sigma_j$, extract contiguous spans of at least 5 tokens, and retain the top 10 segments by peak magnitude. This yields 2,560 segments per dataset (256 modes $\times$ 10 segments). Full extraction statistics are in Appendix D.

Key result. Extracted windows cluster at structural boundaries: problem-to-solution onset, answer emission, follow-up handoff, and prompt reset, instead of at positions encoding semantic content.

5.4.3 Step 3: Feature Labeling

We now ask *what* each mode represents. An LLM-based labeling pipeline assigns each mode a natural-language label, a functional *kind* tag, and a confidence rating. All 256 modes (255 for MATH500) were labeled across all three datasets.

Table 3: Label *kind* distribution. Computational-transitional labels (most reasoning-like) are rare ($< 3\%$).

Kind	GSM8K		MATH500		WikiText-103	
	n	%	n	%	n	%
Event-specific	110	43.0	209	82.0	166	64.8
Structural-rhythmic	107	41.8	41	16.1	79	30.9
Non-discriminable	36	14.1	1	0.4	4	1.6
Computational-transitional	3	1.2	4	1.6	7	2.7

Key result. Labels are overwhelmingly *event-specific* or *structural-rhythmic* (Table 3); only 1–3% are *computational-transitional*. MATH500 is the most event-specific (82.0%), reflecting its templated LaTeX formatting. Labeling confidence is 79.3% (GSM8K), 94.5% (MATH500), 59.4% (WikiText-103) at the high level. Representative labels and confidence breakdowns are in Appendix D.

5.4.4 Step 4: Probing Validation

We then test the internal consistency of the feature labels by training linear probes on the raw latent $\phi(H_k)$ and eigenbasis $c(k) = V^{-1}\phi(H_k)$, predicting binary mode-activation targets (balanced accuracy, chance = 0.5).

Key result. Math-domain trajectories are highly decodable ($> 82\%$ of targets above 0.9 BA), while WikiText-103’s diverse structure yields only 28.7% (Table 4). The near-zero Δ indicates the spectral reorganization does not concentrate information into a cleaner subset; the eigenbasis value lies in temporal factorization rather than improved decodability. Per-mode probing results for representative GSM8K modes are in Appendix E.

Dataset	# Modes	Mean BA		Targets >0.9		$\bar{\Delta}$
		Raw	Eigenbasis	Raw	Eigenbasis	
GSM8K	256	0.948	0.946	83.6%	82.0%	−0.001
MATH500	258	0.945	0.943	82.2%	81.4%	−0.003
WikiText-103	254	0.710	0.711	28.7%	28.3%	+0.001

Table 4: Aggregate linear probing results (balanced accuracy, BA). Raw-latent and eigenbasis probes yield near-identical performance; $\bar{\Delta}$ is the mean eigenbasis–raw difference. **Targets >0.9**: fraction of mode targets whose probe exceeds 0.9 BA (chance = 0.5).

5.4.5 Step 5: Causal Ablation

Steps 1–4 establish correlational evidence: modes activate at interpretable positions and are linearly recoverable. We now test whether modes are *causally* relevant to model output via two tiers of intervention.

Tier 1: Residual-stream perturbation mapping. For each mode, we zero its eigenco-efficient $c_j(k)$, decode back to Hankel space, and measure the resulting perturbation $\Delta h = \psi(z_k) - \psi(z_k^{\setminus j})$. We then compare $\|\Delta h\|_2$ at positions where the mode is active (*on-event*) versus inactive (*off-event*), using energy-matched random modes as a null baseline.

Table 5: Tier 1: residual-stream perturbation from zeroing individual modes. **On/Off**: ratio of mean $\|\Delta h\|$ at on-event vs. off-event positions (values > 1 indicate the mode’s perturbation is larger where it is active). **Significance**: fraction of modes exceeding the energy-matched null at $p < 0.05$ ($z > 1.96$).

Dataset	# Modes	On $\ \Delta h\ $	Off $\ \Delta h\ $	On/Off	Significance
GSM8K	82	84.59	41.77	1.37×	45.1%
MATH500	79	156.77	56.51	2.48×	59.5%
WikiText-103	80	95.60	40.07	1.92×	47.5%

Key result. Approximately half of tested modes show statistically significant event selectivity (Table 5). MATH500 is strongest (2.48×, 59.5% significant), consistent with its templated structure. Perturbations are spatially concentrated: for GSM8K modes 32/33 (answer finalization), the top 5 of 5,120 residual dimensions account for >30% of the L_2 norm (Appendix F).

Tier 2: Logit-shift injection. Tier 1 shows that modes produce spatially selective perturbations; Tier 2 tests whether those perturbations actually shift model predictions. We inject Δh into the LLM’s residual stream at layer 47 and measure the KL divergence between the original and perturbed next-token distributions. Three conditions are compared at scale = 2.0: (i) the mode’s own perturbation direction (**Target**), (ii) a random direction of equal norm (**Random**), and (iii) the perturbation from an energy-matched control mode (**Control**). A high Target/Random ratio indicates the mode encodes a specific direction in output space, not merely a large-norm perturbation.

Table 6: Tier 2: KL divergence between original and perturbed next-token distributions after injecting Δh at on-event positions (scale = 2.0). **Target/Random**: median ratio of mode-specific KL to random-direction KL (values $\gg 1$ confirm directional specificity).

Dataset	# Modes	Target KL	Random KL	Control KL	Target/Random
GSM8K	82	1.001	0.112	1.035	12.58×
MATH500	79	2.398	1.414	1.976	4.73×
WikiText-103	80	1.886	0.384	2.234	6.04×

Key result. Mode-specific perturbation directions produce $4.7\text{--}12.6\times$ larger KL shifts than random directions of equal norm (Table 6), confirming that the modes encode direction-specific information about the output distribution rather than acting as generic noise. On/off selectivity is moderate (median $1.3\text{--}3.4\times$; Appendix F), and the strongest effects ($\text{KL} > 10$) concentrate in a small number of high-energy modes.

5.5 Cross-Architecture Consistency

All three findings—invariant eigenspectra, resolution scaling, and the structural–transient hierarchy—replicate across Qwen2.5-14B-Instruct, DeepSeek-R1-Distill-Qwen-14B, and Mistral-3-14B (Appendix C).

6 Related Work

Dynamical systems views of neural networks. Deep Equilibrium Models (Bai et al., 2019), Neural ODEs (Chen et al., 2018), and Markov chain formulations of LLMs (Zekri et al., 2024; Geng et al., 2026) established dynamical-systems analyses of neural computation. The LSD framework (Anonymous, 2026) applied Koopman operators directly to LLM activation trajectories, recovering spectral signatures invisible in token streams.

Spectral and frequency-domain analyses. HSAD (Li et al., 2025a) applies FFT across layers for hallucination detection; Akrouf (2023) apply DMD to sentence embeddings; NerVE (Jha & Reagen, 2026) tracks feed-forward eigenspectra across depth. These operate on the depth axis or output embeddings; ours analyzes the *temporal* axis of latent activations, learning nonlinear observables.

Koopman operators and deep autoencoders. Koopman theory (Koopman, 1931; Mezić, 2005) lifts nonlinear dynamics into linear observable space. Deep Koopman approaches (Lusch et al., 2018; Aswani & Jabari, 2025) learn neural observable dictionaries; we adapt this architecture to LLM residual streams.

Sparse autoencoders for interpretability. SAEs learn overcomplete feature dictionaries from static activations (Wang et al., 2024). Our pipeline draws direct methodological parallels but operates on *temporal eigenmodes* defined by oscillation frequency and decay rate.

7 Discussion

Main findings. Overall, the KAEs uncover spectral features that are robust across models and datasets. Two findings emerge from spectral analysis: an *invariant eigenspectrum* whose three-component geometry persists across domains, and a *resolution scaling* phenomenon in which increasing latent dimension reveals finer transient structure while preserving previously recovered modes. Both are currently qualitative, and future work should quantify them using coordinate-free measures such as the Wasserstein distance between eigenspectra. Furthermore, the interpretability pipeline reveals that generation dynamics decompose into a *dominant structural skeleton* (formatting, boundaries, turn-taking) and *sparse, low-energy transients* where reasoning-specific computation occurs.

Limitations. First, although the eigenspectra are visually alike, each KAE learns its own observable basis, making cross-dataset numerical comparison ill-posed without shared encoders. Second, probe targets are derived from max-activating windows, making probing a consistency check rather than independent validation. Third, while causal injection shows that mode-specific perturbations shift output distributions far more than random controls, the current framework cannot establish whether these modes play a necessary role in generation or merely correlate with it.

Conclusion and future directions. The results strengthens a dynamical-systems foundation for interpretability of language model latent trajectories. Future directions include shared-encoder architectures for cross-regime comparison, scaling to larger models, and extending the framework to alignment-relevant behaviors such as strategic deception. The KAEs may further be used in other tasks such as speculative decoding, practical execution of which can lead to significant benefits, including but not limited to efficient inference.

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A KAE Training Details

A.1 Hyperparameters

All KAEs share the architecture and training configuration in Table 7. Training uses AdamW ($\text{lr} = 10^{-3}$, weight decay 10^{-5}) for up to 50 epochs (30 for WikiText-103) with early stopping (patience 20). Loss coefficients: reconstruction = 1.0, linear dynamics = 0.1, multi-step rollout = 1.0 with uniform weighting across horizons 1 through N .

Table 7: Common KAE hyperparameters.

Parameter	Value
LLM backbone	Qwen2.5-14B-Instruct
Residual-stream layer	47 (final)
Hankel window (W)	8
Input dimension (Wd)	40,960
Latent dimension (r)	256 (default); sweep {64, 128, 256, 512}
Hidden dimension	1,024
Rollout horizon (N)	8
Batch size	256

A.2 Training Convergence

Table 8 summarizes convergence across datasets. All models reach spectral radii $\rho(K) \approx 1.00$ – 1.01 , indicating near-marginal stability of the learned operator.

Table 8: KAE training summary ($r = 256$, $W = 8$).

Dataset	Epochs	Traj.	Train Loss	Val Loss	$\rho(K)$
GSM8K	50	499	0.839	0.991	1.003
MATH500	50	500	1.014	1.323	1.002
WikiText-103	30	500	1.029	1.135	1.006

A.3 Spectral Diagnostics

We verify that the learned K is diagonalizable by checking $\kappa(V) < 10^3$; all datasets satisfy this threshold (Table 9), so standard eigendecomposition is used throughout.

Table 9: Spectral diagnostics of the learned Koopman operator.

Dataset	$\rho(K)$	$\kappa(V)$	Decomposition
GSM8K	1.003	194.5	Eigen
MATH500	1.002	277.2	Eigen
WikiText-103	1.006	327.5	Eigen

A.4 Comparison to EDMD Baseline

On the truthful/untruthful regime of Anonymous (2026), the nonlinear KAE ($r = 64$, $W = 5$) achieves 1-step test MSE of 0.525 (untruthful) and 0.636 (truthful), degrading to 0.871 and 0.896 at $n = 8$. The coordinate EDMD baseline yields relative MSE of 0.22→0.75 (untruthful) and 0.24→0.79 (truthful)—a much steeper degradation. Because the two methods use different coordinate systems, absolute values are not directly comparable, but the stability gap consistently favors the KAE.

A.5 Cross-Architecture Rollout MSE

Tables 10–11 confirm that the KAE fits generalize to other model families with comparable degradation profiles.

Table 10: KAE rollout MSE on GSM8K (Mistral-3-14B, $r = 256$, $W = 8$). Standardized Hankel space; splits 397/49/50.

Horizon	Train	Val	Test
1-step	0.586	0.656	0.658
4-step	0.597	0.722	0.724
8-step	0.658	0.881	0.887

Table 11: KAE rollout MSE on GSM8K (DeepSeek-R1-Distill-Qwen-14B, $r = 256$, $W = 8$). Standardized Hankel space; splits 400/50/50.

Horizon	Train	Val	Test
1-step	0.565	0.668	0.633
4-step	0.580	0.737	0.700
8-step	0.661	0.898	0.865

B Eigenmode Taxonomy Details

B.1 Energy Fraction and Classification

For each mode j , the energy fraction E_j is computed as the cross-trajectory mean of the temporal variance of $|c_j(k)|^2$, normalized so that $\sum_j E_j = 1$. Let $q_{0.05}$ and $q_{0.50}$ denote the 5th percentile and median of $\{E_j\}$. Modes are classified in priority order:

1. **Transient:** $|\lambda_j| < 0.5$ or $E_j \leq q_{0.05}$.
2. **Backbone:** remaining modes with $|\lambda_j| \geq 0.9$, $|\theta_j| \leq 0.05$, and $E_j \geq q_{0.50}$.
3. **Oscillatory:** all remaining modes.

B.2 Eigenvalue Magnitude Ranges

Table 12 confirms clear separation between classes: backbone modes cluster near the unit circle, oscillatory modes span $|\lambda| \approx 0.50$ – 0.98 , and transients decay rapidly.

Table 12: Eigenvalue magnitude ranges $[\min |\lambda|, \max |\lambda|]$ per class at $r = 256$.

Class	GSM8K	MATH500	WikiText-103
Backbone	[0.926, 1.003]	[0.941, 1.002]	[0.920, 1.006]
Oscillatory	[0.505, 0.976]	[0.502, 0.973]	[0.501, 0.958]
Transient	[0.061, 0.710]	[0.025, 0.648]	[0.022, 0.489]

C Additional Eigenspectra

C.1 Cross-Architecture Spectra (GSM8K)

The three-component spectral geometry persists across model families (Figure 4).

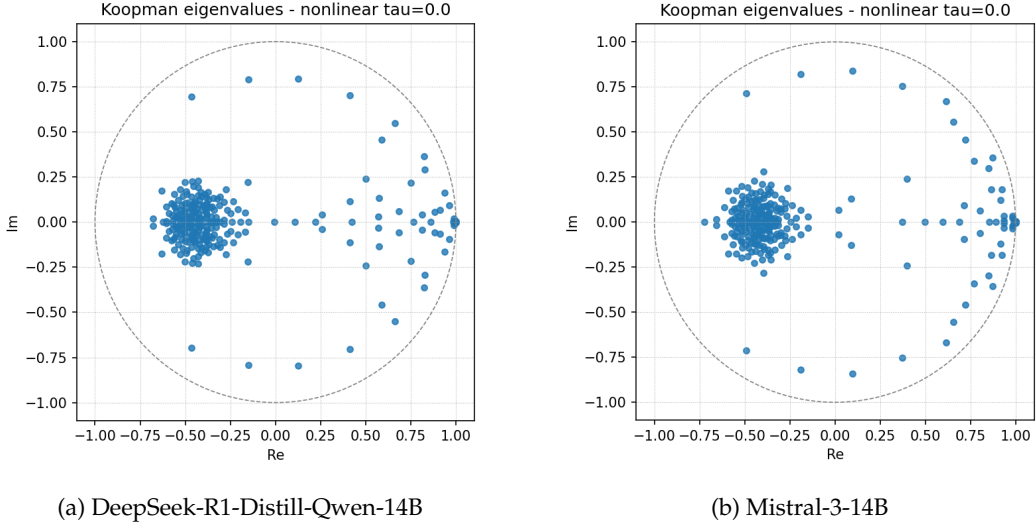


Figure 4: GSM8K eigenspectra at $r = 256$, $W = 8$ for additional model families.

C.2 Resolution Scaling (MATH500)

As with GSM8K (Figure 3), increasing r on MATH500 progressively populates the negative-real cluster while preserving the backbone.

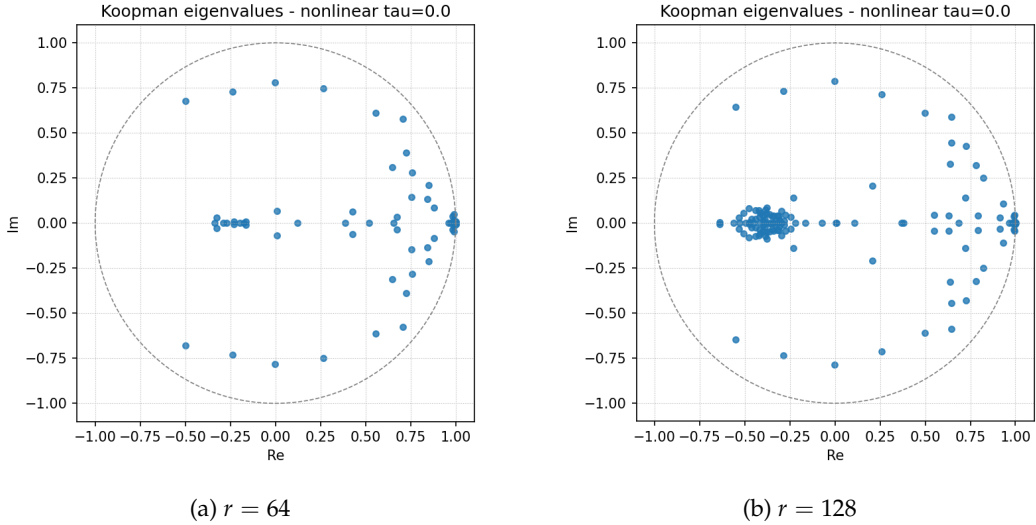


Figure 5: Resolution scaling on MATH500 (Qwen2.5-14B-Instruct, $W = 8$).

C.3 High-Resolution Spectrum (WikiText-103)

At $r = 512$, the WikiText-103 spectrum fills in additional complex-conjugate pairs while the backbone remains stable (Figure 6).

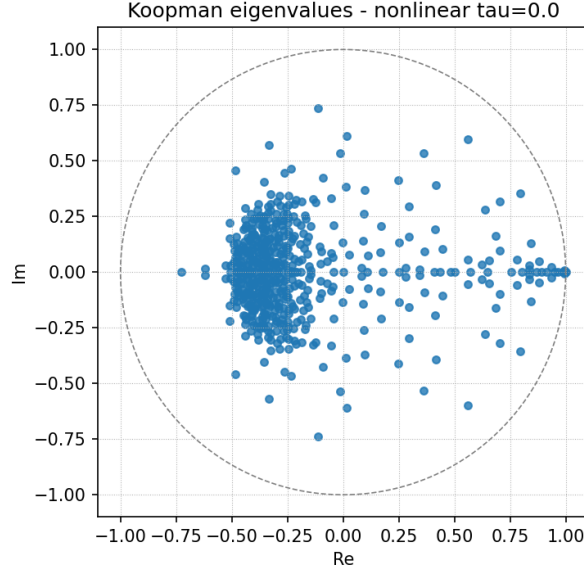


Figure 6: WikiText-103 eigenspectrum at $r = 512$, $W = 8$ (Qwen2.5-14B-Instruct).

C.4 Taxonomized Spectra

Figure 7 overlays the taxonomy from Section 5.4 onto the eigenspectra, showing that backbone (blue), oscillatory (green), and transient (red) classes occupy distinct regions across all datasets.

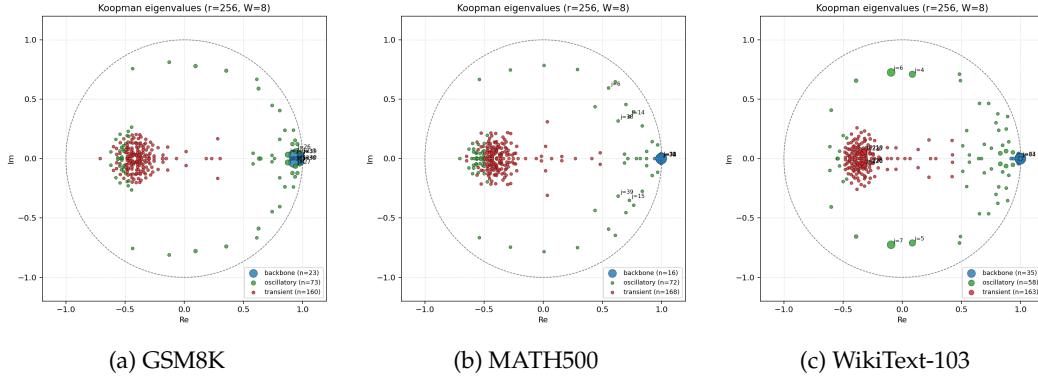


Figure 7: Taxonomized eigenspectra at $r = 256$, $W = 8$ (Qwen2.5-14B-Instruct). Blue: backbone; green: oscillatory; red: transient.

D Interpretability Pipeline Details

This appendix covers max-activation extraction (Step 2) and feature labeling (Step 3) from Section 5.4.

D.1 Extraction Statistics

For each mode, we extract the top 10 segments where $|c_j(k)|$ exceeds $\mu_j + 1.5\sigma_j$ for at least 5 consecutive tokens. Table 13 summarizes the yield on GSM8K; candidate counts vary widely across modes, reflecting differences in activation sparsity.

Table 13: Max-activation extraction summary (GSM8K, $r = 256$, $W = 8$).

Quantity	Value
Modes scanned	256
Stored segments (10 per mode)	2,560
Candidate runs per mode	82–1,470 (median 271)

D.2 Labeling Confidence

An LLM-based pipeline labels each mode with a natural-language description, a functional *kind* tag, and a confidence rating (high / medium / low). Confidence is highest on MATH500 (94.5% high), whose templated LaTeX structure makes modes easy to interpret, and lowest on WikiText-103 (59.4% high), where document diversity introduces ambiguity.

Table 14: Labeling confidence distribution across datasets.

Confidence	GSM8K		MATH500		WikiText-103	
	n	%	n	%	n	%
High	203	79.3	241	94.5	152	59.4
Medium	16	6.2	12	4.7	101	39.5
Low	37	14.5	2	0.8	3	1.2

D.3 Representative Mode Labels

Tables 15–16 illustrate the range of mode functions uncovered by the pipeline. Most modes track structural or event-level boundaries; the rare computational-transitional modes capture reasoning-specific pivots.

Table 15: Representative GSM8K modes from the interpretability pipeline.

Mode(s)	Class	Type	Max-Activation Pattern
90	Transient	Procedural	Follow-up-question boundary; windows near handoff markers and answer-boundary delimiters.
32/33	Backbone pair	Procedural	Final-answer production and transition to next prompt; peaks around boxed numeric answers.
47/48	Backbone pair	Mixed	Problem-to-solution boundary; peaks near end of word problem and onset of solution text.
51/52	Backbone pair	Mixed	Question-boundary and solution-onset detection at delimiter tokens.
10/11	Control pair	Semantic	Surface-template activation tied to repeated patterns (e.g., “be denoted”).

Table 16: Representative labels across datasets.

Dataset	Class	Kind	Label
GSM8K	Oscillatory	Comp.-trans.	Mid-computation pivot: formula application to numerical result extraction
MATH500	Transient	Event-specific	LaTeX/boxed-answer transition onset
MATH500	Transient	Event-specific	New-problem ingestion at prompt boundary
WikiText-103	Backbone	Structural	Document delimiter and passage-QA handoff
WikiText-103	Transient	Event-specific	Script/encoding anomaly detection

E Probing Details

Linear probes are logistic regression (L_2 regularization, $C = 1.0$) trained on 80% of tokens from training trajectories and evaluated on held-out test trajectories. Each mode defines a binary target: 1 at high-activation positions, 0 elsewhere.

Table 17: Per-mode probing for representative GSM8K modes ($r = 256$). Chance = 0.500.

Mode(s)	Target	Raw BA	Eigen BA	Pos.
90	Follow-up boundary	0.977	0.977	298
32/33	Answer finalization	0.772	0.753	134
47/48	Problem-to-solution boundary	0.977	0.974	417
51/52	Question-boundary / solution onset	0.997	0.997	369
10/11	Surface template ("be denoted")	0.999	0.999	10

Boundary and solution-onset modes (90, 47/48, 51/52) are linearly recoverable with BA > 0.97 . The answer-finalization pair 32/33 is weaker (BA ≈ 0.77) due to sparser positives. The control pair 10/11 reaches near-perfect BA on only 10 positives, reflecting a narrow surface-template event.

Table 18: Full probing statistics across datasets.

Dataset	#	Raw Space				Eigenbasis Space			
		Mean	Med.	Min	Max	Mean	Med.	Min	Max
GSM8K	256	.948	.977	.558	.999	.946	.977	.529	.999
MATH500	258	.945	.993	.477	1.00	.943	.993	.477	1.00
WikiText-103	254	.710	.653	.489	1.00	.711	.658	.489	1.00

F Causal Ablation Details

F.1 Tier 1: Perturbation z-Scores

For each mode, we compare the on-event perturbation norm $\|\Delta h\|$ against an energy-matched null distribution (1,000 random modes of comparable E_j). Table 19 reports the resulting z-scores. The heavy-tailed maxima (up to 1,880) indicate that a small number of modes produce perturbations far exceeding chance.

Table 19: Tier 1 z-scores: on-event $\|\Delta h\|$ vs. energy-matched null.

Dataset	Mean	Median	Max	$z > 1.96$
GSM8K	15.32	1.51	512.54	45.1%
MATH500	74.79	3.18	1,880.46	59.5%
WikiText-103	17.01	1.89	897.95	47.5%

F.2 Tier 2: Logit-Shift Injection

Table 20 lists the injection configuration. For each mode, we inject Δh at layer 47 under three conditions (target direction, random direction of equal norm, energy-matched control mode) at both on-event and off-event positions.

Table 20: Tier 2 injection configuration.

Parameter	Value
Target LLM	Qwen2.5-14B-Instruct
Injection layer	47
Scales	{0.5, 1.0, 2.0}
Positions per group	30
Conditions	on/off $\times$ {target, random, control}

Tables 21–22 report KL divergence at scale = 2.0. The large gap between mean and median KL (Table 21) confirms that strong causal effects are concentrated in a small number of high-energy modes. The on/off ratio (Table 22) is moderate, indicating that perturbation effects are not perfectly localized to active positions, while the target/random ratio confirms strong directional specificity.

Table 21: Tier 2 KL divergence from target-direction injection (scale = 2.0, on-event).

Dataset	# Modes	Mean KL	Median KL	Max KL
GSM8K	82	1.001	0.005	13.174
MATH500	79	2.398	0.072	42.968
WikiText-103	80	1.886	0.377	12.007

Table 22: Tier 2 on-event KL by condition and on/off selectivity (scale = 2.0). Target/Random $\gg$ 1 confirms directional specificity; On/Off $>$ 1 indicates partial event localization.

Dataset	On-Event KL			On/Off Ratio	
	Target	Random	Control	Mean	Median
GSM8K	1.001	0.112	1.035	5.07	3.40
MATH500	2.398	1.414	1.976	2.44	1.32
WikiText-103	1.886	0.384	2.234	2.19	1.69

G Cross-Dataset Pipeline Summary

Table 23 consolidates all pipeline metrics for comparison across datasets. The pattern is consistent: a small backbone captures the majority of energy, probing confirms label quality on math domains, and causal injection shows strong directional specificity across all three.

Table 23: Cross-dataset pipeline summary ($r = 256$, $W = 8$, Qwen2.5-14B-Instruct).

Metric	GSM8K	MATH500	WikiText-103
<i>Training</i>			
$\rho(K) / \kappa(V)$	1.003 / 194.5	1.002 / 277.2	1.006 / 327.5
Val loss	0.991	1.323	1.135
<i>Taxonomy</i>			
Backbone / Osc. / Trans.	23 / 73 / 160	16 / 72 / 168	35 / 58 / 163
Backbone energy	56.7%	99.0%	48.6%
<i>Labeling</i>			
Event-spec. / Structural	43.0 / 41.8%	82.0 / 16.1%	64.8 / 30.9%
Comp.-transitional	1.2%	1.6%	2.7%
<i>Probing</i>			
Mean BA (raw)	0.948	0.945	0.710
Eig-raw Δ	-0.001	-0.003	+0.001
<i>Tier 1 Ablation</i>			
On/off ratio	1.37×	2.48×	1.92×
Signif. ($z > 1.96$)	45.1%	59.5%	47.5%
<i>Tier 2 Injection</i>			
Median target/random	12.58	4.73	6.04
Median on/off ratio	3.40	1.32	1.69